

Nested Quantifiers

$\forall x \exists y P(x, y)$: The proposition is same as $\forall x Q(x)$, where $Q(x)$ is $\exists y P(x, y)$.

Let the universe of discourse for the set x and y be the set of positive integers and let $P(x, y): x^2 = y$. Then translate $\forall x \exists y P(x, y)$ in english.

↓

for all x there exists a y such that $x^2 = y$.

E.g *

(1) Translate the given statement into English.

$$\forall x \forall y ((x > 0) \wedge (y < 0) \rightarrow (xy < 0)),$$

where the domain of both variables consist of all real numbers.

(2) Let $P(x, y)$ be the statement " $x + y = y + x$ ".

What are the truth values of the quantifications $\forall x \forall y P(x, y)$ and $\forall y \forall x P(x, y)$, where the domain for all variables consist of all real numbers.

(3) Let $Q(x, y)$ denote " $x + y = 0$ ". What are the truth values of $\exists y \forall x Q(x, y)$ and $\forall x \exists y Q(x, y)$, where the domain of all variables consist of all real numbers.

TABLE 1 Quantifications of Two Variables.

<i>Statement</i>	<i>When True?</i>	<i>When False?</i>
$\forall x \forall y P(x, y)$ $\forall y \forall x P(x, y)$	$P(x, y)$ is true for every pair x, y .	There is a pair x, y for which $P(x, y)$ is false.
$\forall x \exists y P(x, y)$	For every x there is a y for which $P(x, y)$ is true.	There is an x such that $P(x, y)$ is false for every y .
$\exists x \forall y P(x, y)$	There is an x for which $P(x, y)$ is true for every y .	For every x there is a y for which $P(x, y)$ is false.
$\exists x \exists y P(x, y)$ $\exists y \exists x P(x, y)$	There is a pair x, y for which $P(x, y)$ is true.	$P(x, y)$ is false for every pair x, y .

Date

e.g:- Translate the statement "The sum of two positive integers is always positive" into a logical expression.

sol:- Let us introduce two variables x and y .
Both $x, y \in I^+$

We can rewrite the given statement as
"for all positive integers x and y , $x+y$ is positive",
which can be expressed as

$$\forall x \forall y ((x > 0) \wedge (y > 0) \rightarrow (x+y > 0))$$

OR

"for every two positive integers, the sum of these integers is positive".

$$\forall x \forall y (x+y > 0)$$

where the domain for both variables consists of all positive integers.

Date

e.g.:- Translate the statement "Every real number except zero has a multiplicative inverse."

Sol.:- We first re-write this as "for every real number x except zero, x has a multiplicative inverse." We again can re-write this as "for every real number x , if $x \neq 0$, then there exists a real number y such that $xy = 1$." This can be expressed as

$$\forall x ((x \neq 0) \rightarrow \exists y (xy = 1)).$$

e.g.:- Translate the statement

$$\forall x (C(x) \vee \exists y (C(y) \wedge F(x, y)))$$

into English, where $C(x)$ is " x has a computer," $F(x, y)$ is " x and y are friends" and the domain for both x and y consists of all students in your school.

Sol. The statement says that for every student x in your school, x has a computer or there is a student y such that y has a computer and x and y are friends.

OR

Every student in your school has a computer has a friend who has a computer.